PY-STEM · ACTIVITY PYS-08 · MECHANICS

Low-Ropes Force Map Relay

Predict balance and stability before the ropes element proves you right.

Win condition: Most accurate predictions and clear debrief connections.

THE SCIENCE YOU ARE USING

Center of mass. Your center of mass is the average position of your weight. You stay balanced while it stays over your base of support. Lean too far and the line of gravity leaves the base, so you tip. Lowering your center of mass adds stability.

Mapping forces. On each ropes element, you predict where forces act and where balance will be hard, then test the prediction with your body and a partner. Accurate predictions and a clear debrief score.

HOW TO BUILD AND RUN IT

1. **Read the element.** Look at the next low-ropes element and the body positions it requires.
2. **Predict balance points.** On the card, mark where balance will be hardest and where to put your weight.
3. **Test on the element.** Follow host rules and try it. Notice where you actually felt unstable.
4. **Map the forces.** Sketch where forces acted and how you kept your center of mass over your base.
5. **Debrief the connection.** Explain how lowering or shifting your center of mass changed stability.

Safety: Host procedures override this module. This is a wraparound only. Allergy: Outdoor sensitivity note.

MATERIALS

Element cards	set
Clipboards	6
Center-of-mass templates	per team
Pencils	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why):

The ONE variable we are testing:

Baseline result:

After redesign:

DEFEND YOUR DESIGN

What evidence made you change your design?

If you ran it again, what is the ONE thing you would change?

HOW YOU SCORE (100 POINTS)

Scoring criterion	Points
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	**100**